

FIN Local Research — Release 10.42

Checkpoint P458–P464

Unique Palindromic Curvature, Fiberwise Canonical Allocation, and a
Thirty-Fold Dual-Gap Reduction

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1 August 2026

Publication — Preprint; Version 1.0.0

Central result. A new exact-rational nested-dual certificate proves

$$0.52332810026048937 \leq D_3^{\text{global}} \leq 0.52332832027103,$$

reducing the certified gap to $2.2001054063 \times 10^{-7}$.

Additional theorems. The declared palindromic coarse-erasure maximizer is unique and isolated to width below 1.22×10^{-13} . Under the explicit minimum-TV rule, every supplied detector-box fiber has one unique strictly convex minimax allocation.

Boundary. No exact full-cone optimizer, full-simplex uniqueness, selector, dimensional source, laboratory record, complete legacy–strict bridge, role transfer, or physical closure is claimed.

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Publication statement

This monograph reports local analytical and computer-assisted Programs P458, P459, and P464. It uses no laboratory record, external audit, internet result, remote computation, or physical validation.

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Reproduction certificate

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Chapter 1

Executive summary

This checkpoint completes local Programs P458, P459, and P464 by analytical derivation, rational arithmetic, outward interval automatic differentiation, exact finite linear algebra, and deterministic local computation. It uses no laboratory record, external audit, internet result, remote computation, or physical evidence.

1. **P458** — **[Computer-assisted proof]**. The P452 symmetrization theorem and P457 global cover reduce the declared coarse-erasure question to a bounded palindromic maximizer hull. On 256 outward interval cells,

$$-4.002571478626325 \leq F''(a) \leq -3.890264298788524 < 0.$$

Endpoint derivatives have opposite certified signs. Consequently the palindromic maximizer is unique and lies in

$$0.2276921883225489 \leq a_* \leq 0.22769218832267057,$$

a width of 1.2169×10^{-13} . This proves uniqueness of the palindromic representative, not the absence of asymmetric co-optimizers.

2. **P459** — **[Proven, conditional]**. P453 fixes one signed atomic representation under the explicit minimum-TV rule. For every realization of the supplied O160/O163 detector coefficient box, the risk

$$R(q) = \sum_{i=0}^6 \frac{s_i^2}{q_i} - M$$

is strictly convex on the probability simplex and has the unique solution

$$q_i = \frac{s_i}{\sum_j s_j}.$$

The smallest certified ambient Hessian diagonal is greater than 90.5081, and all adjacent allocation ranks remain separated by at least 0.00589644924709949. The result is fiberwise canonical; it does not choose an actual detector realization or provide calibration.

3. **P464** — **[Computer-assisted proof]**. A smaller-barrier nested-dual scout was admitted only after rounding to denominator 10^{14} , exact Sylvester verification of six matrix slacks, positivity of two scalar slacks, and payment of the complete trigonometric perturbation. The new global full-cone bracket is

$$0.52332810026048937 \leq D_3^{\text{global}} \leq 0.52332832027103.$$

Its exact width is

$$\frac{22001054063}{10^{17}} = 2.2001054063 \times 10^{-7},$$

approximately 29.999 times narrower than the Release-10.41 bracket. The corresponding success probability lies in

$$0.761664050130244685 \leq p_{\text{succ}}^{\text{global}} \leq 0.761664160135515.$$

New objects are O173 Palindromic Curvature Isolator, O174 Fiberwise Canonical Allocation Tube, and O175 Rational-Dual Admission Gate.

No exact P464 optimizer, full-simplex uniqueness, selector, canonical physical orientation, dimensional unit, apparatus, laboratory record, complete legacy-to-strict bridge, role-transfer theorem, Standard Model, gravity, L_{total} , or Theory of Everything is claimed.

Chapter 2

Confidence convention

Label	Meaning
[Proven]	Exact analytic or finite algebraic consequence of declared premises
[Computer-assisted proof]	Finite rational or outward-interval certificate with auditable artifacts
[Strong evidence]	Reproducible numerical evidence lacking a complete certificate
[Conditional]	The conclusion depends on an explicitly supplied mathematical interface
[Open]	Current local artifacts do not decide the proposition
[Blocked by external evidence]	Laboratory, calibration, custody, or an external record is indispensable

Chapter 3

State map and guardrails

Lane	Inherited state	New state
P457 maximizer	Global value gap below 10^{-6} ; uniqueness open	Unique palindromic maximizer isolated to width $< 1.22 \times 10^{-13}$
O163 allocation	Certified interval tube	Unique minimizer in every supplied-box fiber; rank order invariant
Three-slot full cone	Gap $6.59999151063 \times 10^{-6}$	Gap $2.2001054063 \times 10^{-7}$
Exact O167 attainment	Open	Open
Full-simplex uniqueness	Open	Open: asymmetric equality cases not excluded
Selector/QW-2191	Open	Unchanged
Dimensional source	Missing	Unchanged
Laboratory evidence	None	None
Legacy-to-strict bridge	Incomplete	Unchanged

The intermediate and strict kernels remain distinct:

$$K_{\text{legacy,ont}}(d) = \frac{\alpha_{\text{geo}} \cos(\omega d + \phi)}{1 + \beta_{\text{tors}} d}, \quad K_{\text{strict,gate}}(d) = \frac{\cos(\omega d + \phi)}{1 + \beta d^n}.$$

These programs are downstream reduced-channel and estimator calculations. They derive neither kernel and transfer no legacy physical role. The ontology remains nadsoliton to light to matter to emergent observer, with no lower informational layer beneath the nadsoliton.

Chapter 4

P458 — interval curvature and unique palindromic maximizer

4.1 Proof architecture

P452 proves that reversal averaging cannot reduce the declared coarse objective, hence a global optimizer has a palindromic representative

$$p(a) = (a, \tfrac{1}{2} - a, \tfrac{1}{2} - a, a).$$

P457 proves that every palindromic maximizer is contained in

$$H = [0.2178955078125, 0.234619140625].$$

P458 differentiates the same block-reduced objective, including every square root, Pfaffian absolute value, and standard-sector contribution, with second-order outward interval jets. No finite difference is used in the proof layer.

4.2 Strict curvature

The 256-cell cover of H proves a strictly negative second derivative on every cell. At the hull endpoints,

$$F'(H_-) \in [0.03881422991489784, 0.03881422991492505],$$

$$F'(H_+) \in [-0.027120573501572463, -0.027120573501547406].$$

Strict decrease of F' , continuity, and opposite signs prove exactly one stationary point in H , which is its strict maximum.

4.3 Scope of uniqueness

The theorem is deliberately narrower than full-simplex uniqueness. P452 shows that symmetrization sends every point to a no-worse palindromic point. Equality could still hold along a non-palindromic

fiber. Excluding that possibility requires a strict equality-case theorem for the P452 reversal inequality.

Chapter 5

P459 — fiberwise canonical detector allocation

5.1 Variational theorem

For strictly positive scores s_i , the Hessian of $R(q) = \sum_i s_i^2/q_i - M$ is diagonal:

$$\nabla^2 R(q) = \text{diag} \left(\frac{2s_i^2}{q_i^3} \right) \succ 0.$$

Thus the restriction to the simplex tangent space is strictly positive. Lagrange multipliers give the unique minimizer $q_i = s_i / \sum_j s_j$. Every certified score lower bound is positive over the complete inherited atom box, so the argument holds in every fiber.

5.2 Certified allocation order

The interval image is centered at

$$(0.240000916, 0.234104467, 0.224592449, 0.131228143, 0.080622667, 0.068262231, 0.021189127).$$

All adjacent interval differences are positive. The smallest is greater than 0.00589644924709949, so label ordering cannot change anywhere inside the supplied box.

5.3 Removal test

Removing the minimum-TV premise revives the P450 signed null-cycle and makes the allocation representation-dependent. Retaining minimum-TV but removing a specified detector realization leaves a tube of unique fiberwise answers, not one physical allocation. No preparation or calibration theorem is generated.

Chapter 6

P464 — rational dual admission and global bracket

6.1 Admission rule

Floating central-path values are scouts only. A candidate enters the theorem layer only if every rationalized slack is positive. For the accepted denominator- 10^{14} certificate:

- both $X_3 \pm \Delta/2$ have exact Sylvester margin 10^{-8} ;
- the four lower matrix slacks have exact margin 10^{-8} ;
- both terminal scalar slacks equal $2000001/10^{14}$;
- the trigonometric matrix perturbation is bounded by $3.0514767686602317 \times 10^{-16}$.

Hence the smallest final certified slack exceeds $9.9999996948523228 \times 10^{-9}$.

6.2 Theorem and limitation

Weak duality gives the new upper bound 0.52332832027103. Combined with the P451 rational primal witness it proves the global bracket stated above. This does not prove that the O167 candidate attains the exact optimum: the remaining gap is positive. It also supplies no uniqueness theorem.

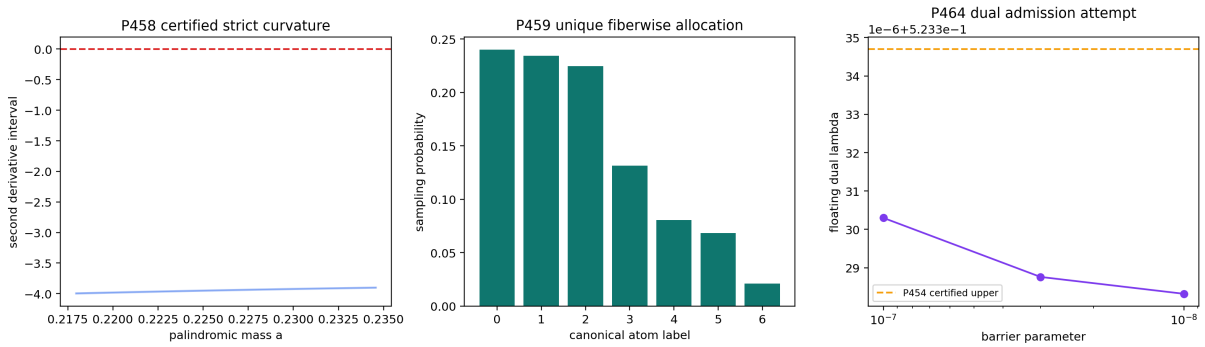


Figure 6.1: Certified palindromic curvature, fiberwise detector allocation, and the admitted nested-dual refinement.

Chapter 7

Newly constructed objects

Object	Definition	What it generates	What it does not generate
O173	Second-order interval jet cover on the P457 hull	Strict curvature and unique palindromic root	Full-simplex equality-case uniqueness
O174	Map from each positive detector-score fiber to its unique simplex minimizer	Canonical allocation conditional on minimum-TV and a detector fiber	Physical detector choice or calibration
O175	Exact rational feasibility gate for nested dual scouts	Admissible global upper bounds only	Exact primal attainment or physical evidence

Chapter 8

Falsification and robustness

- P458 recomputes derivatives through the complete block formula; sampled Hessians are not used as proof.
- P458 does not erase possible asymmetric co-optimizers.
- P459 proves strict convexity rather than inferring uniqueness from one numerical minimization.
- P459 explicitly fails without the P453 minimum-TV gauge.
- P464 rejects floating decreases unless all exact rational slacks pass.
- P464 pays the full certified trigonometric operator perturbation.
- None of the programs changes selector, dimensional, kernel-bridge, or laboratory-evidence status.

Chapter 9

Physical and epistemic boundary

The results are dimensionless mathematical statements about supplied finite operators, probability laws, and synthetic detector coefficients. They do not derive time, length, action, energy, apparatus, a preparation instrument, independent custody, or an experimental record. QW-2191 remains open. No mathematical output in this checkpoint is empirical confirmation of FIN.

Chapter 10

Ranked next research programs

Rank	Program	Question	Method
1	P465	Does P452 have strict equality only on the palindromic fixed set?	Factor and interval-certify the reversal Jensen defect
2	P468	Can the P464 gap be reduced below 10^{-8} ?	Primal KKT inclusion plus adaptive rational dual margins
3	P469	Does an exact stationary O167 witness satisfy full nested complementarity?	Interval matrix sign/KKT system
4	P460	Is the O174 allocation invariant under alternative admissible loss geometries?	Quotient and sensitivity theorem
5	P461	Formalize the P452 symmetrization theorem	Lean finite algebra
6	P462	Formalize P453 strict-complementarity uniqueness	Lean moment-duality core
7	P463	Specify a preparation contract without claiming its physical realization	Typed operational interface
8	P466	Test whether causal coherence advantage persists for four slots	Reduced comb recursion and certified witness
9	P467	Generalize O170 to arbitrary slot number	Inductive primal/dual ladder theorem
10	P470	Identify which local conclusions are stable under strict-kernel parameter boxes	Interval perturbation audit with no role transfer

The next bounded batch is

$$\boxed{P465 \longrightarrow P468 \longrightarrow P469}.$$

Chapter 11

Artifact list and restart

- FIN_Programs_458_459_464_Results.json
- FIN_Programs_458_459_464_Summary.csv
- FIN_Programs_458_459_464_P458_Derivative_Uniqueness.csv
- FIN_Programs_458_459_464_P459_Canonical_Allocation.csv
- FIN_Programs_458_459_464_P464_Dual_Refinement.csv
- FIN_Programs_458_459_464_P464_Rational_Dual.npz
- FIN_Programs_458_459_464_Figures/p458_p459_p464_certificates.png
- fin_programs_458_459_464.py
- test_fin_programs_458_459_464.py

Recompute with:

```
MPLCONFIGDIR=/tmp/fin-mpl-1038 python3 fin_programs_458_459_464.py
MPLCONFIGDIR=/tmp/fin-mpl-1038 python3 -m unittest -v \
    test_fin_programs_458_459_464.py test_fin_checkpoint_p458_p464.py
python3 fin_programs_458_459_464_to_latex.py
lualatex -interaction=nonstopmode -halt-on-error \
    FIN_Local_Research_Checkpoint_P458_P464_EN.tex
sha256sum -c FIN_PROGRAMS_458_459_464_RELEASE_MANIFEST.sha256
```

Chapter 12

Conclusion

The strongest surviving interpretation is mathematical and operational: a finite nested causal optimization admits auditable primal and dual certificates; a declared coarse-erasure model has one rigorously isolated palindromic optimum; and an explicit variational gauge produces a unique detector allocation in every supplied uncertainty fiber. These are substantive local theorems. They still do not manufacture the selector, dimensional scale, physical preparation, or laboratory record required for a physical theory.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/fin-mpl-1038 \  
python3 fin_programs_458_459_464.py  
python3 -m unittest -v \  
    test_fin_programs_458_459_464.py \  
    test_fin_checkpoint_p458_p464.py  
python3 fin_programs_458_459_464_to_latex.py  
lualatex -interaction=nonstopmode -halt-on-error \  
    FIN_Local_Research_Checkpoint_P458_P464_EN.tex  
sha256sum -c FIN_PROGRAMS_458_459_464_RELEASE_MANIFEST.sha256
```

Suggested citation: Żuchowski, K. (2026). *FIN Local Research Checkpoint P458–P464: Unique Palindromic Curvature, Fiberwise Canonical Allocation, and a Thirty-Fold Dual-Gap Reduction* (Release 10.42; Version 1.0.0) [Preprint].